

Class sets and Eisenstein congruences on compact exceptional groups

III. The compact F_4 at level 5

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Abstract

We determine the class set of the compact F_4 at the parahoric level attached to the prime 5 (the stabiliser of a sharp-null six-space in $J/5J$): it has 259 classes, 54 over the Albert lattice $J_{\mathbb{Z}}$ and 205 over the exotic lattice J_E , found by uniform and fixed-point sampling of six-spaces with exact stabilisers and exact isomorphism tests, and certified complete by the mass formula, which is met exactly on both halves, together with the independent identity $\sum n_Z/|\text{Stab}| = 713/18$. The Eisenstein primes of this level are therefore 7, 13, 31, 313, 601 (and 691 through $\Theta(\Delta)$). We describe the sampling computation of the Hecke operator $T(\omega_1^\vee)(2)$ on these classes, including a necessary refinement of the isomorphism test (equivalence of the intersection lattices alone merges distinct classes; the isometry must carry the Albert lattice itself), and we record, before the operator is complete, the eigenvalues predicted for the lifts of the classical newforms of level 5 by the dictionaries of Part II, with the primes each is predicted to carry.

1 The level and the main theorem

Let $J = J_{\mathbb{Z}}$ be the integral Albert algebra of Part II and J_E the second class of its genus, $|\text{Aut}(J_{\mathbb{Z}})| = 4,180,377,600$, $|\text{Aut}(J_E)| = 634,023,936$. A six-dimensional subspace $S \subset J/5J$ on which the sharp map vanishes (a *sharp-null six-space*) determines the lattice $\Lambda_S = G_0(S) + 5J$, with $G_0(S)$ the 21-dimensional space of elements whose cross product with S vanishes; the stabiliser of Λ_S in $F_4(\mathbb{Q}_5)$ is a parahoric subgroup Q_5 , and the class set of the compact F_4 at level Q_5 is the set of pairs (J', S) , J' in the genus of J , modulo $F_4(\mathbb{Q})$, i.e.

$$F_4(\mathbb{Q}) \backslash F_4(\mathbb{A}_f) / K = \text{Aut}(J_{\mathbb{Z}}) \backslash \{S \subset J_{\mathbb{Z}}/5J_{\mathbb{Z}}\} \sqcup \text{Aut}(J_E) \backslash \{S \subset J_E/5J_E\},$$

with 38,208,007,656 six-spaces on each side. The mass formula gives $\sum_{J'} \sum_S 1/|\text{Stab}(S)| = 75809539/8294400$ on the $J_{\mathbb{Z}}$ side (Part II) and 5831503/96768 on the J_E side; a second identity counts, for each S , the number $n_Z(S)$ of its five 5-neighbours isomorphic to $J_{\mathbb{Z}}$: $\sum n_Z/|\text{Stab}| = 713/18$ on the J_E side. Both identities are exact rational numbers, so a list of pairwise distinct orbits whose masses sum to them contains every orbit.

Theorem 1.1. *The class set of the compact F_4 at level Q_5 has 259 classes: 54 over $J_{\mathbb{Z}}$ (Part II) and 205 over J_E . The 205 orbits of $\text{Aut}(J_E)$ on sharp-null six-spaces of $J_E/5J_E$ have stabiliser orders distributed as in Table 1; their masses sum to exactly 5831503/96768 and $\sum n_Z/|\text{Stab}| = 713/18$ exactly. Consequently the Eisenstein primes of this level (the primes of the numerator of the mass, $2^7 \cdot 3^6 \cdot 7^2 \cdot 13^2 \cdot 31^2 \cdot 313 \cdot 601$ times the Bernoulli factor) are 2, 3, 7, 13, 31, 313, 601 and 691.*

Stab	1	2	4	6	8	12	16	18	24	32
orbits (J_E)	11	66	24	39	13	11	5	1	14	4

Stab	54	64	96	108	168	192	336	1536	6048
orbits (J_E)	1	2	6	1	1	2	2	1	1

Table 1: Stabiliser orders of the 205 orbits over J_E . The $J_{\mathbb{Z}}$ side: 8, 14, 4, 1, 7, 4, 1, 1 orbits with stabilisers 2, 4, 6, 8, 12, 24, 36, 48, the twelve large orbits 96, 120, 128, 192^3 , 1152, 2880, 3072^2 , 5184, 9600, and the rare orbit 110,592.

2 Method: finding 205 orbits

Sampling and fingerprints. Uniformly random sharp-null six-spaces (two runs, 41,963 in all) were reduced to the invariants of the intersection lattice $M = \Lambda_S$ (counts of vectors of norm ≤ 4), its stabiliser $|\text{Stab}(S)|$ (the Jordan automorphisms among the automorphisms of M , by the exact test of Part II), and the types and stabilisers of the five lifts. This gives 110 keys, but keys are coarser than orbits: 24 invariant vectors are shared by orbits of different stabilisers, and many keys contain several orbits with the same stabiliser.

Exact splits. For every key, the samples were clustered by the exact isomorphism test (qfisom on the intersection lattices followed by the Jordan test on the resulting isometry), 300 samples per key in two runs (by invariant, then by invariant and stabiliser), and a third run for eleven keys whose automorphism closures made the first runs time out. The hit rate per orbit, 702 ± 70 samples per unit of $|\text{Stab}|$, is uniform enough that every key’s orbit count is pinned; the top key alone holds ten orbits of stabiliser 2.

The rare orbits. The residual mass after the splits was $367/96768 < 1/192$, so the missing orbits have stabilisers in the hundreds or thousands. Fixed-point sampling (six-spaces invariant under representatives of the 49 conjugacy classes of $\text{Aut}(J_E)$, 298,075 candidates with enlarged symmetry) produced exactly three new invariant vectors, with stabilisers 6048 and 1536 computed and the third too large for the closure; since $1/336 + 1/1536 + 1/6048 = 367/96768$ exactly, the third has stabiliser 336 and the list is complete.

3 The Hecke operator $T(\omega_1^\vee)(2)$: method and a lemma

The operator acts on functions on the class set by $f \mapsto \sum f(J', S')$ over the 139,230 two-neighbours J' of J (rank-one points of $J/2J$, lifted by Hensel’s lemma and realised as unimodular Albert lattices in the ambient algebra), with S' the six-space of $\Lambda_S[\frac{1}{2}] \cap J'$. With 259 classes the 36 million exact identifications of a full computation are out of reach, since the invariant keys hold several orbits each and the exact test costs seconds; we estimate the matrix instead by 150 random neighbours per class, identifying each exactly among the 258 representatives, and restore the two exact constraints, row sums 139,230 and the mass symmetry $T_{ij}/|\text{Stab}_j| = T_{ji}/|\text{Stab}_i|$. Both lattices $J_{\mathbb{Z}}$ and J_E sit in the same rational Albert algebra (J_E has basis $L/4$ for an integral matrix L), so identification is intrinsic, in ambient coordinates scaled by 16, after LLL reduction of every basis.

Lemma 3.1. *Two pairs (J, S) , (J', S') are in the same class if and only if some $F_4(\mathbb{Q})$ -isometry carries Λ_S onto $\Lambda_{S'}$ and J onto J' . Equivalence of the lattices Λ_S alone is strictly weaker: an isometry of Λ_S onto $\Lambda_{S'}$ may carry J onto a different Albert lattice containing $\Lambda_{S'}$.*

This was found the hard way: without the second condition the test merged two orbits of stabilisers 16 and 8 sharing an invariant vector, and the ambient “stabiliser” counts 22 and 23 were

not group orders; with it, the pair stabilisers 16 and 8 are reproduced exactly. The sampling runs that built the class set were unaffected, since they compared six-spaces inside a fixed lattice, where the isometry is integral.

4 Predictions

The dictionaries of Part II, re-anchored on 15057, 2331 and 207 at level 3, predict the eigenvalues of $T(\omega_1^\vee)(2)$ on the lifts of the classical newforms of level 5, and the classical Eisenstein congruences of those newforms (verified in their coefficient fields at $q \leq 13$) predict which Eisenstein primes they carry.

Prediction 4.1. The spectrum of $T(\omega_1^\vee)(2)$ at level Q_5 contains 139230 (Eisenstein), 8631 ($\Theta(\Delta)$, carrying 691), 14157.94 and 4868.06 (the $F_4 \times \mathrm{PGL}_2$ lifts of the quadratic pair of weight-12 newforms of level 5, carrying 601), 12285 (the lift of the rational weight-12 newform, $a_2 = 34$, carrying 31), 6668.08 and 1297.92 (the $\mathrm{Sp}_6 \times \mathrm{SL}_2$ lifts of the quadratic pair of weight-8 newforms, carrying 313), 1935 (the 207-type lift of the weight-6 newform, $a_2 = 2$, carrying 31) and 91 (the $\mathrm{Sp}_6 \times \mathrm{SL}_2$ lift of the rational weight-8 newform, $a_2 = -14$, carrying 13). The prime 7 has no classical source in weights 6, 8, 12 and is predicted to fall on a stable form or a G_2 -lift.

The sampled operator (in progress) tests this list; a confirmation would extend the carriers theorem of Part I to a fourth independent level, with the “exceptional” primes (7 here, as $p \mp 1$ on G_2) again on forms with no classical origin.

Data and code

GitHub `chrisyounge-create/Cayley-Agent-Repository: f4p5/` (`pathB_E.py`, `orbit_split_reps_E.py`, `fixsample_E.py`, `classify_hunt_E.py`, `reps_ambient.json`, `nbr_sampler.py`, `identify.py`, `opsample.py`, `assemble_operator.py`); all runs as GitHub workflows with artifacts collected into `results/`.